THE GLUEING FUNCTOR IS FULLY FAITHFUL ON SPACES SEPARATED OVER X

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2392-2404 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2406-2427, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 27 of queue_spaces-pushouts.json (pb=1097, stmt 2392-2404, proof 2406-2427).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AF3: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2319–2321. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AF4: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2326–2340. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AF5: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2376–2379. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AEV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2046–2060. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03KO: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2427–2427. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 0AEV the functor (0AF3) is fully faithful on algebraic spaces separated over X. More precisely, it induces a bijection*

$$\mathrm{Mor}_X(X'_1, X'_2) \longrightarrow \mathrm{Mor}_{Spaces(Y \to X, Z)}(F(X'_1), F(X'_2))$$

whenever $X'_2 \to X$ is separated.

Proof. Since $X'_2 \to X$ is separated, the graph $i : X'_1 \to X'_1 \times_X X'_2$ of a morphism $X'_1 \to X'_2$ over X is a closed immersion, see Morphisms of Spaces, Lemma 03KO. Moreover a closed immersion $i : T \to X'_1 \times_X X'_2$ is the graph of a morphism if and only if $\mathrm{pr}_1 \circ i$ is an isomorphism. The same is true for

(1) the graph of a morphism $U \times_X X'_1 \to U \times_X X'_2$ over U,
(2) the graph of a morphism $V \times_X X'_1 \to V \times_X X'_2$ over V, and
(3) the graph of a morphism $Y \times_X X'_1 \to Y \times_X X'_2$ over Y.

Moreover, if morphisms as in (1), (2), (3) fit together to form a morphism in the category $Spaces(Y \to X, Z)$, then these graphs fit together to give an object of $Spaces(Y \times_X (X'_1 \times_X X'_2) \to X'_1 \times_X X'_2, Z \times_X (X'_1 \times_X X'_2))$ whose triple of morphisms are closed immersions. The proof is finished by applying Lemmas 0AF4 and 0AF5. □